



A Collaborative Filtering Recommendation Algorithm Based on Community Detection and Graph Neural Network

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Abstract

Recommendation system is an important module of many online systems. As one of the mainstream methods in the current recommendation system, the recommendation algorithm based on graph neural network can not only learn the cooperation signal between nodes, but also detect the nonlinear high-order information in node interaction. However, like the traditional collaborative filtering algorithm, the collaborative filtering method based on graph neural network only depends on the interaction information of existing users and projects, and the recommendation effect is poor in the scenario of sparse data. Although incorporating item attributes as auxiliary information into the recommendation algorithm can alleviate this problem to a certain extent, directly using the item attributes that users browse to construct user embedded vectors cannot contain users' implicit preferences. Therefore, this paper constructs a user–user network based on the interaction information between users and projects, analyzes it by using community detection, divides user groups with common interests, mines the potential attributes of users, and designs a collaborative filtering recommendation algorithm based on community detection and graph neural network. The research on two large real data sets shows that our method is superior to the standard recommendation method and the latest collaborative filtering recommendation method based on graph neural network.

Keywords Collaborative filtering · Community detection · Item attributes · Graph neural network

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1 Introduction

The development of the Internet has led to the explosive growth of data. These data have rich value, but they also bring a serious problem of “information overload” [1]. How to quickly and effectively obtain valuable information from complex data has become a key issue in the development of big data research. As an effective method to solve the problem of information overload, recommendation system has become a research hotspot in academia and industry, and has been widely used such as content-based recommendation [2], collaborative filtering recommendation [3] and association rule-based recommendation [4].

Among the recommendation algorithms, collaborative filtering is the most widely used. Collaborative filtering predicts whether users are interested in an item according to the historical evaluation (score) of all users on purchased items. At present, collaborative filtering using graph convolution neural network technology is the focus of current research [5]. By constructing the user item bipartite graph and using the graph convolution network to aggregate the information of users and item nodes, this technology can not only retain the original user item interaction information, but also mine the higher-order implicit relationship between users and items, alleviate the problem that it is difficult to detect the high-order relationship between users and items by traditional methods such as matrix decomposition, and achieve better recommendation effect. The traditional collaborative filtering algorithm does not rely on the attribute information of items or the description of users, but only on the historical behavior of users. Although this is simple and low-cost in data acquisition, it has poor effect in scenes with sparse data. In order to alleviate the problem of sparse data, the existing methods mainly add auxiliary information, as an important auxiliary information, item attribute information has been widely used in recommendation system [6, 7].

In the recommendation system, the embeddings of users are usually determined by the items they browse. Therefore, after the introduction of item attribute features, users can be represented by browsing item attributes. However, although this can represent users’ explicit preferences, it cannot detect users’ implicit preferences.

In complex networks, nodes can be classified by community detection algorithm, and users’ preferences can be explored according to the degree of similarity of nodes. Community is an important attribute of the network. It is regarded as a set of nodes with close connection in the community and loose connection between communities. Community detection is used to reveal the classification of nodes and the division of community structure, which is of great significance for the study of complex networks.

This paper constructs a user–user network through the interaction information between users and items, classifies users through community detection algorithm, excavates users’ implicit preferences, and constructs more comprehensive user embeddings. The graph convolution network is used to construct the recommendation model, and the item attribute characteristics and user characteristics are combined with it in the way of weight, so as to improve the recommendation accuracy of the algorithm.

To sum up, the main work of this paper is as follows:

1. It proposes a new way to build a user-user relationship network through the interactive information between users and items. By exploring the interactive distribution law of users and items on the real data set, this paper constructs a user-user weighted relationship network, which can more clearly observe the personality and characteristics of users.
2. It proposes a more comprehensive representation of user attribute features. This paper introduces item attributes as auxiliary information, and takes the characteristics of items

- browsed by users as the explicit attribute characteristics of users; For the network constructed in 1, different user groups are divided by efficient community detection algorithm, and the attribute features contained in the group are taken as the implicit features of users. The combination of the two constructs a more comprehensive user attribute feature.
3. Combined with the attribute characteristics of users and items, it designs a collaborative filtering recommendation algorithm CD-GCN based on community detection and graph neural network, and verifies its accuracy on real datasets.

2 Related Work

Modern recommendation system [8] represents users and items through vectors, and recommends items to users according to the similarity between user vectors and item vectors. Among them, collaborative filtering method (CF) is the most widely used. For example, MF [9] decomposes the adjacency matrix constructed by user and item interaction into user matrix and item matrix, and inner product them to predict interaction. In order to make the embeddings of users and items more accurate, many studies use the way of adding auxiliary information to model design, such as adding item content [10], social relations [11], item relations [12], user comments [13], etc.

As an important side information, attribute features have been widely used in recommendation system, but the effect is often poor of directly constructing attribute features as the vector representation of goods or users as the input of model. Because neural network is usually divided into multiple layers. When important features are input from the bottom layer and the graph convolution layer is stacked deeply, the nodes will be too similar and much information will be lost, which is called “excessive smoothing” phenomenon in graph neural networks. There are usually two ways to alleviate this problem. One is to add features in the part close to the output layer, in other words, add features shallower, such as wide & deep [14], which put important attribute features into the wide part; The other method is to convert the important attribute features into weights and multiply them with the user and item embeddings, so that the features can act as judges. For example, LHUC [15] structure converts attribute features into weights with the same dimension as the input vector, multiplies them with the vector, and then takes them as the input of the upper neural network layer. Although the second method can effectively enhance the role of important attribute features in the model, it is difficult to explore the high-order features of users when building the user embeddings through the user’s historical behavior.

Although the traditional direct inner product prediction can express the explicit preference of users, it still belongs to the linear model, so it is impossible to explore the high-order features in user project interaction [16]. Therefore, the current research focuses more on the use of deep learning technology to enhance the interaction function [17], so as to capture the nonlinear interaction between users and projects. For example, the collaborative filtering model adopts the way of neural network, such as NeuMF [18], which uses the way of neural network instead of inner product to represent the implicit preference of users.

In addition to the deep learning method, because the interactive information between users and items can build a bipartite graph, the graph neural network can also be applied to collaborative filtering recommendation. For example, GC-MC [19] applies graph convolution network (GCN) [20] to user item bipartite graph, but it only uses one convolution layer, so it cannot reveal the collaborative information of user items in high-order connectivity. NGCF [21] adopts multi-layer convolution layer for learning, so that the vector representation of

users and items can contain higher-order collaborative information, but it adopts nonlinear activation function and feature transformation, which makes the model structure more redundant. Light-GCN [22] has been improved on NGCF to remove redundant transformation operations and improve the effect.

Community structure is an important feature in complex networks. Through community detection, we can find the structure and information hidden in the network and the potential relationship between nodes. According to the different community structure, community detection algorithm can be divided into overlapping community detection and non-overlapping community detection, and also can be divided into dynamic network and static network according to the network structure. Now there has been a lot of relevant research. In the non-overlapping static community detection method, Louvain [23] proposed an algorithm based on modularity optimization. The algorithm performs well in time complexity and accuracy, and can find hierarchical community structure. Rosvall et al. Proposed the Infomap [24] method in 2008. This method is based on node coding and random walk, and the effect is better in large-scale networks. In addition, there are many related studies, such as LPA [25], which is based on label propagation method; walktrap [26] based on random walk, etc. Through community detection, users' hidden attributes can be mined. Through community detection, we can mine the hidden attributes of users. In the recommendation algorithm, we can use this method to mine the potential attributes of users and further expand the features of users.

3 Model

3.1 Model Overview

As shown in Fig. 1, this paper refers to the graph convolution technology, use GCN to learn the representation of users and items, and introduces item attributes as side information to improve the effect of the algorithm.

In the GCN model, the initial user embeddings e_u^0 and item embeddings e_i^0 as the input, the three-layer graph convolution neural network is used to learn the high-order features of the interaction between users and items, and then the embedding of each order are spliced to obtain the representation of users and items.

After getting the vector representation of users and items through graph convolution technology, this paper introduces attributes as auxiliary information. The specific way is to express attributes as weight vectors (W_u, W_i) with the same dimensions as user node and item node vectors, and multiply them bit by bit with node vectors to enhance their vector representation. Attribute weight construction is divided into two aspects: user attribute weight and item attribute weight. The attribute weight of the project is obtained by dimension reduction through the attribute tag of the project, and the construction of the attribute weight of the user is divided into two parts: the user's own attribute and the user's community attribute. The user's own attributes are obtained through the attributes of the item that the user browses, while the user's community attributes are found on the user user interaction network built based on the user and project information to find the user's community and get the attributes contained in the user's community. Finally, the two parts are weighted and summed to obtain a more comprehensive user attribute weight vector, which alleviates the problem that it is difficult to learn user potential feature information by directly introducing attributes in the way of feature weight.

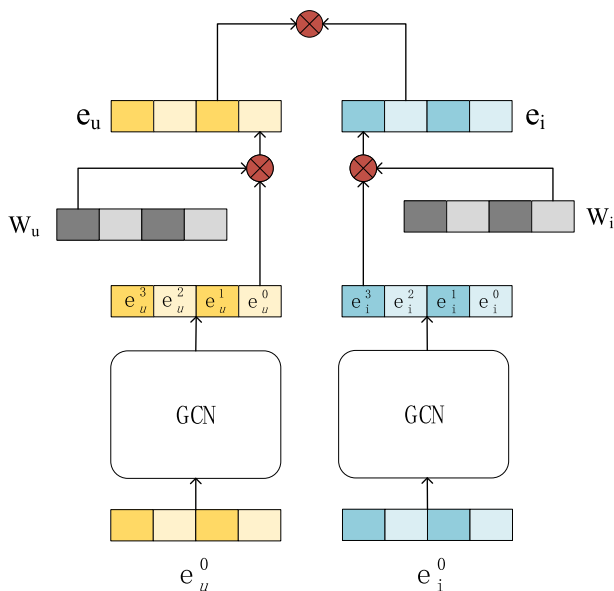


Fig. 1 CD-GCN model

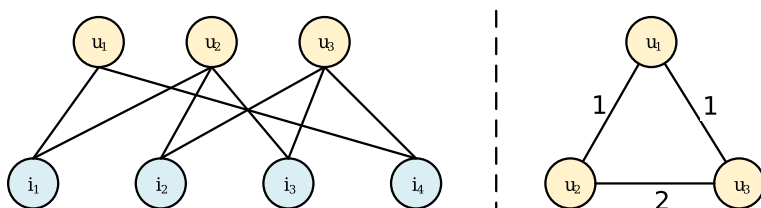


Fig. 2 User-User interaction graph

After obtaining the enhanced vector representation of users and items, perform the inner product operation on them, and judge the similarity between the two vectors according to the inner product result. The higher the similarity, the easier the user clicks the item. The TopK sequence recommended to users is obtained according to the similarity ranking, so as to achieve the purpose of recommendation.

3.2 User-User Interaction Graph

For the original data set with users and items, it is easy to build a user item bipartite graph, in which users and items are each part of the bipartite graph. If there is interaction between user nodes and item nodes, there is an edge between the two nodes.

Although the user item binary graph can build a collaborative filtering and recommendation network of graph neural network, when the interaction between users and projects is sparse, the constructed bipartite graph can not capture the relationship between users very well. Therefore, community division can be used to mine hidden relationships between users as auxiliary information to enhance the feature vector representation of users. Therefore, as shown in Fig. 2, this paper constructs a user-user weighted interaction graph according to the

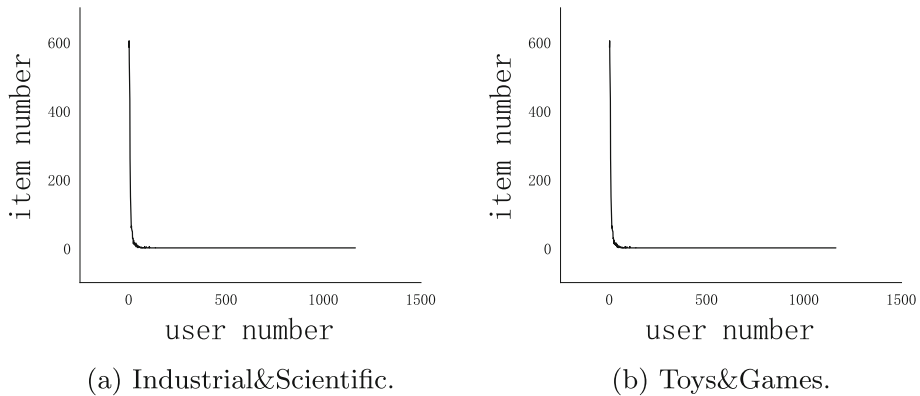


Fig. 3 Distribution of the number of items interacting on different real datasets

interaction information between users and items.

$$W(u_a, u_b) = \sum_{k \in S} E(u_a, i_k) \cap E(u_b, i_k) \quad (1)$$

$$S = i_{u_a} \cup i_{u_b} \quad (2)$$

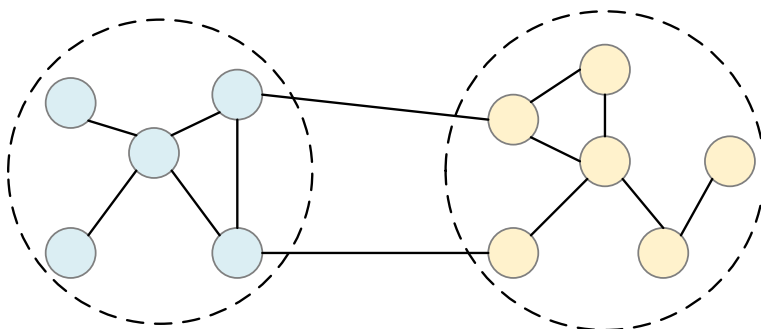
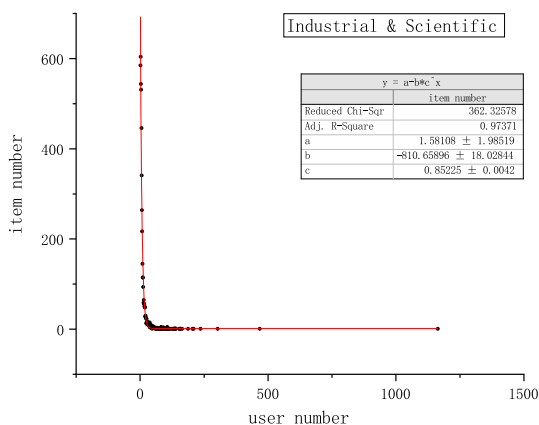
As shown in Formula (1) and Formula (2), W is the edge weight between user nodes. If the value of W is 0, it means that there is no edge between two users. If the value of W is not 0, it means that there is edge between two users, and the weight is W . Where u means the user node, i means the item node, and E is the interaction between the user and the item. If there is interaction, the value is 1, otherwise, the value is 0. S represents the collection of items interacted by u_a and u_b .

Formula (1, 2) can build a basic user relationship graph, but in practice, there is a defect in directly establishing the edges according to the common interaction of users. Generally, there are some items that have been interacted by a large number of users in the dataset, such as browsing ordinary daily necessities like paper drawing in the e-commerce dataset or selling books in the book purchase dataset. These items are very universal, and a large number of users will interact with them, which will lead to the construction of the network can not clearly include the differences of different users. For example, user A and user B have interacted with tissue, so they are connected. In fact, we can't think they are the same kind of people by interacting with toilet paper.

Therefore, we explore the distribution of the number of items interacted on different real recommendation data sets, and optimize the user relationship graph.

Figure 3 shows the interactive distribution of items on Amazon Toys&Games dataset and Amazon Industrial&Scientific dataset. The x-axis indicates how many users have interacted with the items, and the y-axis indicates how many such items there are. It can be seen that the two datasets have almost the same distribution law in the interactive distribution of items, which is that more than 90% of the items are interacted by a small number of users, while a small number of items are interacted by a large number of users, and the distribution forms a long tail trend.

As shown in Fig. 4, taking the Industrial&Scientific dataset as an example, we gradually fit the part after the number of interactions begins to decline. It can be obtained that if $x > 100$, the function value is basically close to the constant, that is, there are very few items interacted by users greater than 100. Through the above exploration, this paper divides

Fig. 4 distribution fitting curve of Industrial&Scientific dataset**Fig. 5** Community structure

the construction of user relationship network into two steps. First, remove the interactive information of items interacted by a large number of users (determined according to the distribution fitting function), and then construct the network through formula (1, 2), which can effectively reduce the complexity of the network, and can more clearly observe the user's personality and characteristics.

3.3 Community Detection

In order to better explore the potential attributes of users from the user-user relation graph constructed in the previous section, this paper explores the experimental effects of classical community detection algorithms on the network, including LPA algorithm based on label propagation, worktrap algorithm and Infomap algorithm based on random walk, Louvain algorithm and Newman [27] algorithm based on modular optimization. Figure 5 shows the network structure after community detection. The division effect is measured by modularity, the number of divided communities and the standard deviation of community size distribution. The main indicator to measure the effect of community division is modularity, which represents the tightness of the community's internal structure and the looseness of external relations.

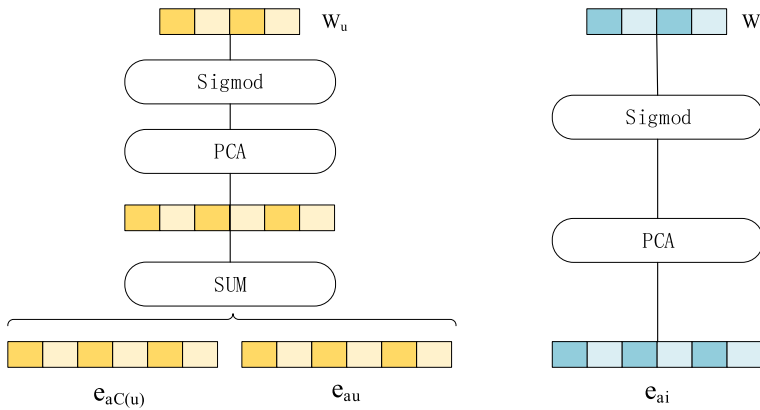


Fig. 6 Weight of attribute

$$Q = \frac{1}{2m} \sum_{i,j} \left[a_{ij} - \frac{d_i d_j}{2m} \right] \sigma(C_i, C_j) \quad (3)$$

Formula (3) is the calculation method of modularity, where m is the total number of edges in the network, a_{ij} is the connection of nodes i and j . if they are directly connected, the value of a_{ij} is 1, otherwise it is 0. d_i and d_j represent the degree of node i and node j , and C_i and C_j respectively represent the community to which nodes i and j belong. The function $\sigma(C_i, C_j)$ indicates whether i and j belong to the same community, if $C_i = C_j$, the function value is 1, otherwise it is 0.

In addition, if there are too few communities or too many users in each community, a monster community will be formed, which will lead to serious homogenization of users and make it difficult to observe the potential personality and characteristics of users. Therefore, we also take the number of communities and the standard deviation of community size distribution as the measurement standard.

As shown in Table 1, from a comprehensive point of view, Infomap algorithm combines information coding and information entropy to analyze the network, which is better than the label propagation algorithm, and has finer division granularity than the method of directly taking modularity as the optimization function. Therefore, Infomap has a better division effect in such large networks, while maintaining a high degree of modularity, the number of community nodes is more evenly distributed.

To sum up, this paper uses Infomap algorithm to divide the user-user interaction graph into communities and explore the potential preferences of users.

3.4 Weight of Attribute Features

As shown in Fig. 6, Building the weights of attribute features is mainly divided into two parts: user side feature weight and item side feature weight. Formula (4, 5 and 6) are the initialization method of user and item attribute vectors, during initialization, the onehot method is used to encode the attribute vector of the items; the attribute vector of user is derived from the attributes of the items they have interacted with; the attribute vector of the community to which the user belongs is derived from the attribute vectors of all the users it contains.

Table 1 Performance of community detection algorithms on real datasets

Datasets	Index	LPA	Louvain	Worktrap	Newman	Infomap
Toys and games	Modularity	0.0033	0.4161	0.3443	0.3581	0.393
	Community numbers	3	14	50	4	198
	Standard deviation	9097.16	1732.81	1093.78	2243.15	455.84
Industrial and Scientific	Modularity	0.2706	0.5829	0.5349	0.4338	0.546
	Community numbers	36	32	197	39	359
	Standard deviation	1629.67	444.46	333.69	451.37	70.08

$$e_{ai} = \text{onehot}(l_i) \quad (4)$$

$$e_{au} = \sum_{i \in I_u} e_{ai} \quad (5)$$

$$e_{aC(u)} = \sum_{u \in C(u)} e_{au} \quad (6)$$

Where e_{ai} represents the attribute vector of item i , l_i represents the attribute tag of item i , e_{au} represents the attribute vector of user u , I_u is the set of articles browsed by user u , $e_{aC(u)}$ represents the attribute vector of the community to which user u belongs, and $C(u)$ is the community of user u .

$$e_{au} = w_1 e_{au} + w_2 e_{aC(u)} \quad (7)$$

The next step is to combine the user attribute vector and the user community attribute vector to form a new user attribute vector. As shown in formula (7), w_1 and w_2 are weight parameters. The weight of the user's own attribute information should be greater than the weight of the user's potential attributes according to the community detection. In this paper, it is set as $w_1 : w_2 = 4 : 1$, which has a good effect.

Because there are many kinds of attribute tags, the dimension of the encoded vector will be large. In this paper, principal component analysis (PCA) is used to reduce the dimension of the vector. This method can highlight the main characteristics of the vector and make it have a fixed dimension. Finally, sigmoid method is used to map the vector value to 0–1 to obtain the weight of user's attribute features. For the item side, the same method is used to obtain the weight of item's attribute features.

3.5 GCN Model

According to the idea of message construction and message aggregation in GCN and the experimental exploration in light-GCN [22], there are redundant operations in the direct use of GCN in message transmission, which will bring adverse effects. Therefore, the feature transformation and nonlinear activation feature are abandoned in the process of message construction and aggregation.

$$e_i^{(k+1)} = \sum_{u \in N_i} \frac{1}{\sqrt{d_i d_u}} e_u^{(k)} \quad (8)$$

$$e_u^{(k+1)} = \sum_{i \in N_u} \frac{1}{\sqrt{d_i d_u}} e_i^{(k)} \quad (9)$$

Where e_u is the vector representation of u (user node), e_i is the vector representation of i (item node). $N_i(N_u)$ indicates the neighbor node of node $i(u)$. d_i and d_u are the degrees of nodes i and u , $\frac{1}{\sqrt{d_i d_u}}$ is the message attenuation factor, $e_u^{(k)}$ is the k -order aggregation represented by the embedding of user node u , and $e_i^{(k)}$ is the k -order aggregation represented by the embedding vector of item node i .

$$e_u = W_u \sum_{i=0}^k w_k e_u^k \quad (10)$$

$$e_i = W_i \sum_{i=0}^k w_k e_i^k \quad (11)$$

In this model, the aggregation method of each order embedding of user u and item i is as show in formula (10, 11). Where k is the number of aggregation layers, w_k is the parameter of the weight of each layer of aggregation in the last vector, which is generally set as $1/(k+1)$. W_u is the weight of user's attribute features and weight of item's attribute features. Finally, the embeddings of users and items are inner product to calculate their similarity, and finally get the recommended candidate set.

3.6 BPR Loss Function

Bayesian Personalized Sorting (BPR) is a Pairwise method, which is used to solve the recommendation sorting problem of implicit feedback, such as clicking, browsing, purchasing and other interactive behaviors that can not clearly reflect users' preferences.

BPR uses Pairwise's idea to build a partial order relationship. It does not simply regard non feedback data as negative examples, but combines with positive examples to build a partial order relationship. The core assumption here is that a user's preference for items he has feedback on must be higher than that of items he has not feedback on. Items that have not feedback include real negative examples and missing values. BPR attempts to build a complete partial order relationship for each user through the user's feedback matrix. The formula is as follows:

$$L_{BPR} = - \sum_{u=1}^M \sum_{i \in N_u} \sum_{j \notin N_u} \ln \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|E^{(0)}\|^2 \quad (12)$$

4 Experiment

This section introduces the selected dataset, and explores the impact of different ways of building user user interaction networks on the network structure and community discovery effect. After that, it introduces two indicators for evaluating the recommendation algorithm and related baseline models. Finally, this chapter explores the impact of different community algorithms and the proportion of community attributes in user attribute rights on CD-GCN, evaluates the effect of CD-GCN on two real data sets, and compares it with other baseline models.

4.1 Datasets

In order to evaluate the effectiveness of the experiment, we use two sub datasets in Amazon public dataset [28], which has been widely used in the recommendation system. And we use 5-core dataset, and each user has interacted with at least five items. In the experiment, the ratio of training set to test set is 8:2, which means each user has at least 4 historical behaviors for training and 1 historical behavior for testing. Table 2 provides a detailed description of the experimental data sets used in this chapter.

4.2 User–User Network Construction

Table 3 is a comparison of the number of user user interaction network edges constructed by removing hot item interactions and retaining hot item interactions. It can be seen that after

Table 2 Number of users, items, interactions, attributes and network density of each dataset

Datasets	User	Item	Interaction	Density	Tags
Industrial & scientific	11,024	5334	65,982	0.00,112	1024
Toys & games	19,412	11,924	167,596	0.00,072	412

Table 3 Comparison of the number of edges between two network construction methods

Datasets	Original network	Lightweight network
Industrial & scientific	839,008	598,074
Toys & games	1,676,322	1,341,131

removing hot item interaction information, the number of network edges is significantly reduced by about 20–25%, and this part of hot items only accounts for a small number of all items. For example, only three of the 5334 items in the Industrial and Scientific dataset are classified as hot items, Only 32 of the 11,924 items in Toys and Games are classified as hot data.

Figure 7 shows the division effect of different construction methods of user user interaction network in Toys and Games dataset and Industrial and Scientific dataset in different communities. Since the LPA algorithm has a relatively general performance effect on various indicators and has little reference value, three community discoveries, infomap, louvain and walktrap, are selected for comparison. Orange light represents a lightweight user user interaction network constructed after removing hot items, while green origin represents a user user interaction network constructed directly using the original method.

It can be seen that in Toys and Games and Industrial and Scientific data sets, lightweight network construction has more communities divided in infomap, Louvain and walktrap algorithms, and the distribution variance of the number of nodes in the community is smaller, which shows that the user user interaction network constructed after removing the interactive information of hot projects can better explore the user's personality characteristics.

4.3 Evaluating Indicator

In this paper, recall and NDCG, which are commonly used in the recommendation system, are used to evaluate the experiment.

Recall rate refers to the proportion of the number of correctly predicted samples in the total number of real samples, focusing on the proportion of successful prediction in real samples. It is defined as follows:

$$recall@k = \frac{TP}{TP + FN} \quad (13)$$

TP (True Positive) means that positive samples are predicted as positive samples, that is, the predicted result items also appear in the real user browsing sequence; FN (False Negative) means that positive samples are predicted to be negative samples, that is, the predicted result items do not appear in the user's real browsing sequence.

The normalized NDCG consists of two parts: DCG and IDCG. DCG is the combination version of cumulative gain (CG) and position weight. IDCG refers to the maximum value of DCG under ideal conditions. The formula for combining the two is as follows:

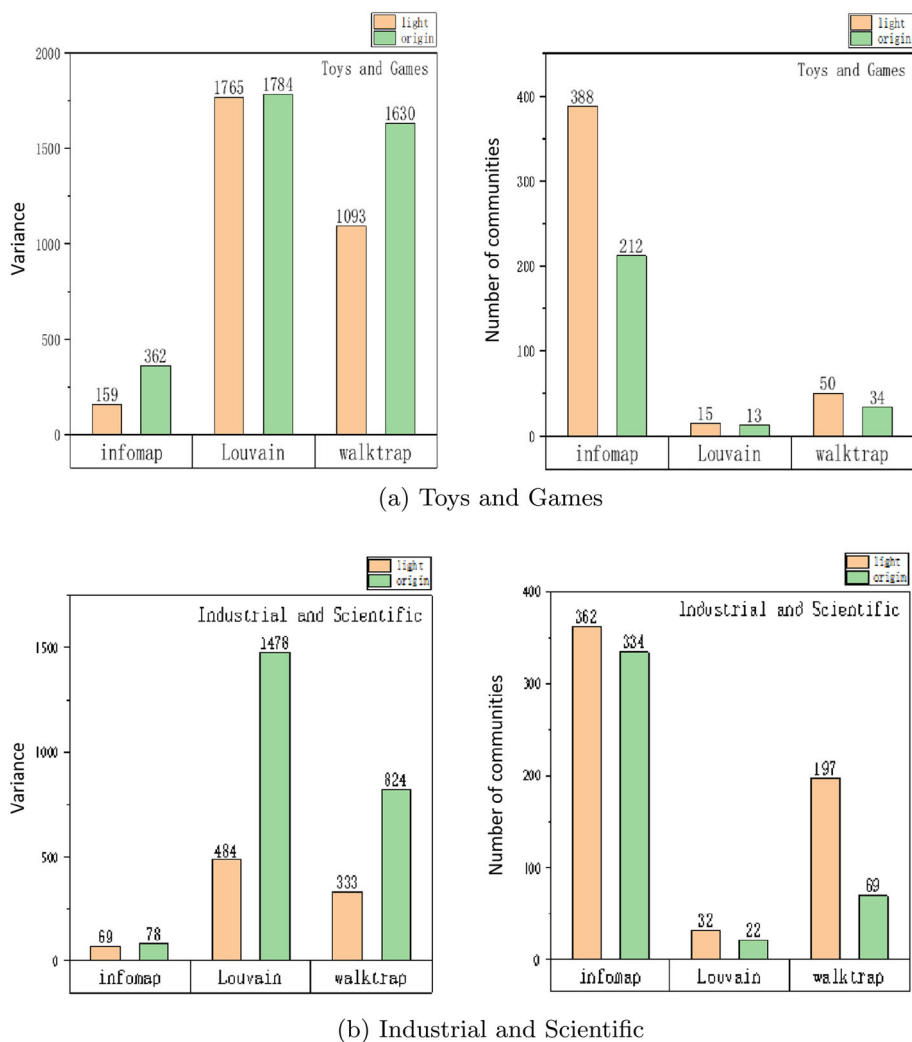


Fig. 7 Community Discovery Effects on Two Networks

$$NDCG@k = \frac{1}{k} \sum_{j=1}^k \frac{1}{\log_2(I_j + 1)} \quad (14)$$

where k is the first k items of the candidate set, I_j represents the location of the index of the j th prediction. Compared with recall, the NDCG considers the order attribute of the recommended candidate set. The higher the index of the prediction item that hits the result, the higher the score. In this paper, $k=20$ is used to evaluate the experimental results.

4.4 Baseline Model

In order to effectively evaluate the effect of CD-GCN recommendation, this paper selects the following baseline experiment based on graph neural network for comparison:

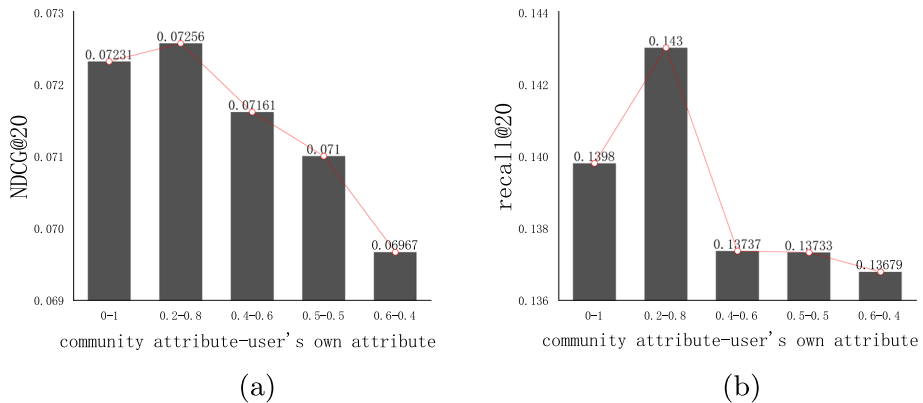


Fig. 8 Impact of the proportion of users' own attributes and community attributes on the effect of recommendation model

GC-MC the model uses GCN [18] coding to generate the vector representation of users and projects, and uses the form of graph automatic coding for collaborative filtering recommendation. This method only considers the first-order neighbors. Therefore, a layer of map convolution is adopted.

NGCF the model adopts the idea of GCN message aggregation and message construction, regards the interaction between users and projects as a bipartite graph, learns the vector representation of nodes through message construction, enables them to aggregate the information of neighbor nodes, and then aggregates the information of nodes at all levels through message aggregation, enables them to represent the high-order information of nodes, and excavates potential interaction relationships.

Light-GCN this model is an optimized version of NGCF model. After exploring that many operations in the message construction part of NGCF model are unnecessary, such as nonlinear feature transformation, it removes useless operations in NGCF and makes it lighter and more effective.

4.5 Experiment Setting

The experimental setting is mainly divided into two parts: parameter setting and environment configuration.

Environment configuration: this experiment adopts tensorflow 1.14 framework. The GPU is NVIDIA GTX2060.

Parameter setting: the embedding size of the vector is uniformly set to 64, the batch size is fixed to 1024, the dropout rate is set to 0.1, and the learning rate is set to 0.001. Except that GC-MC model adopts one-layer graph convolution network layer, other models adopt three-layer graph convolution. In addition, the model is evaluated every 10 epochs, if the recall index does not increase within 20 evaluations, the experiment will be ended in advance.

Figure 8 shows the performance of our recommendation model on the Toys and Games dataset by using the Infomap community detection algorithm. According to the different proportion of the user's own attribute and the community attribute, it can be seen that no matter the evaluation index is recall or NDCG, when the weight of community attributes and

Table 4 Performance of each algorithm in Amazon datasets

Datasets	Index	GC-MC	NGCF	Light-GCN	CD-GCN	Improve
Industrial & Scientific	recall@20	0.04170	0.05903	0.06627	0.06821	2.92%
	NDCG@20	0.02332	0.03530	0.03884	0.04014	3.35%
Toys& Games	Recall@20	0.06159	0.12362	0.13526	0.14306	5.77%
	ndcg@20	0.02881	0.06302	0.06953	0.07253	4.31%

users' attributes is 0.2 and 0.8, that is, when the proportion is 1:4, the effect of the model is the best. Increasing the weight of community attributes will reduce the effect of the algorithm.

4.6 Experimental Result

Proportion of the user's own attribute and the community attribute In Sect. 3.4, we weighted the user's own attribute weight w_1 and the community attribute weight w_2 to distinguish the difference between the influence of the user's explicit attribute and potential attribute on predicting the user's choice, and carried out the corresponding experiments based on this. In this paper, the commonly used evaluation indexes in the recommendation system, recall and normalized impairment cumulative gain (NDCG), are used to evaluate the experiment.

Such experimental results show that although community detection can find the potential attributes of users, it can not think that all attributes in the community must be the potential attributes of users. In fact, the importance of explicit attributes should be greater than potential attributes. Therefore, the experimental results are better when the weight of community attributes is less than the weight of users' own attributes. After the above experimental exploration, we set the weights of community attributes and users' own attributes to 0.2 and 0.8 to achieve better calculation results.

Comparison with baseline experiment Table 4 shows the experimental results of CD-GCN algorithm and baseline algorithms on Industrial and Scientific dataset and Toys and Games dataset. The bold style in the table represents the best effect. Among the baseline algorithms based on graph neural network, Light-GCN has the best effect, which improves the performance by more than 40% compared with GC-MC and more than 10% compared with NGCF algorithm. In the Industrial and Scientific dataset, the CD-GCN algorithm proposed in this paper is compared with the Light-GCN algorithm in recall@20 and NDCG@20, it increased by 2.92% and 3.35% respectively, and increased by 5.77% and 4.31% in the Toys and Games dataset. To sum up, the CD-GCN proposed in this paper can achieve good recommendation effect.

Ablation experiment As shown in Fig. 9, we conducted ablation research on CD-GCN to explore the impact of community division and project attribute enhancement matrix on the recommendation effect, and implemented the following two simplified CD-GCN variants:

- CD-GCN-c, remove the user community attribute matrix W_u .
- CD-GCN-i, remove the item attribute matrix W_i .

Based on the ablation experiment, we concluded that:

- (1) Adding user community attribute matrix or project feature attribute matrix separately will significantly improve model performance
- (2) As a whole, the user community attribute matrix and the project feature attribute matrix have a very positive impact on the model at the same time.

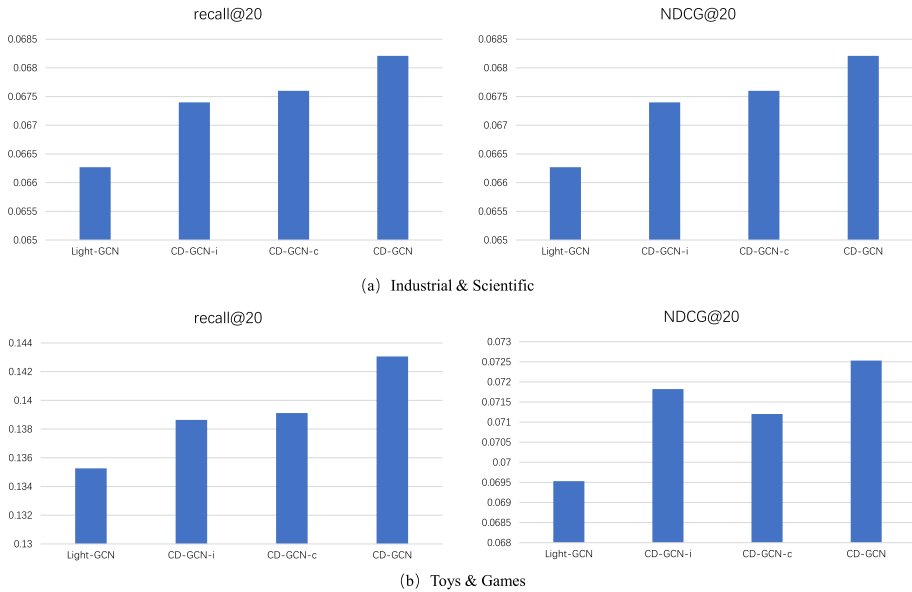


Fig. 9 Ablation experiment

5 Conclusion

This paper proposes a collaborative filtering recommendation algorithm based on community detection and graph neural network, constructs user attributes through item side attributes, mines the relationship between users through the interactive data set of users and items, detects users' potential attributes by using community detection method, and introduces GCN model for collaborative filtering recommendation, It has achieved good experimental results in the real network data set.

6 Discussion

The CD-GCN model proposed in this paper has achieved good performance in the test of actual data sets, but there are still some shortcomings: first, the user user interaction network built in CD-GCN is an undirected weighted graph, on which a non overlapping static weighted undirected graph community discovery algorithm is used, but in fact, users may belong to several different hobby circles, Therefore, it may be better to use overlapping community discovery algorithm to explore the potential attributes of users. Secondly, the weight ratio adjustment of community attributes and user attributes can be changed from empirical conclusions to model adaptive learning.

Author Contributions Conceptualization, JS and QL: methodology, QL: supervision, JS and BW: validation, QL and ZH: writing, JS and QL: All authors have read and agreed to the published version of the manuscript.

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Declarations

Conflict of interest The authors declare no conflict of interest.

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